Substance Abuse Screening and Treatment

Johnny C. Tenegra, MD, MS*, Bobby Leebold, LCSW

KEYWORDS

• Substance abuse disorder • Substance misuse • Screening • Substitution therapies
• Blocking therapies

KEY POINTS

- Although the prevalence of substance use disorders in primary care patients runs high, physicians, due to lack of adequate training and skepticism about treatment effectiveness, feel unprepared to tackle substance abuse disorders in the office.
- Diagnosis of a substance abuse disorder can be made through a combination of history taking, screening questionnaires, physical examination, and chemical testing.
- Primary care physicians are well-positioned to treat those with substance abuse with knowledge and access to behavioral techniques, community resources, and pharmacologic strategies.
- The correlation of substance use disorders with other medical and psychiatric comorbidities makes the treatment of substance use disorders all the more important to patient care.

INTRODUCTION

The role of a primary care physician in the management of substance use disorders can be complex and challenging. The scope of the problem is significant and underscores a public health epidemic. In 2013, an estimated 24.6 million Americans aged 12 or older were current illicit drug users, representing 9.4% of the population.[1] In the general population, roughly 15.5 million were dependent on or abused alcohol alone, with some 100,000 people dying each year in the United States as a result.[2] Furthermore, in 2004, it was estimated that more than 9.4% of Americans older than the age of 12 had a full-blown drug or alcohol addiction.[3]

The issue is made even more complicated by the increase in the nonmedicinal use of both prescription and over-the-counter (OTC) medications. This problem is particularly prevalent among adolescents, in whom the use of these drugs has steadily escalated over the past few years. The most common drugs of choice include stimulants, sedatives, tranquilizers, and most notably, specific analgesics such as opioids.

Decatur Family Medicine Residency, Southern Illinois University School of Medicine, 102 West Kenwood Avenue, Suite 100, Decatur, IL 62526, USA
* Corresponding author.
E-mail address: jtenegra@siumed.edu

Prim Care Clin Office Pract 43 (2016) 217–227
http://dx.doi.org/10.1016/j.pop.2016.01.008 primarycare.theclinics.com
0095-4543/16/$ – see front matter

The abuse of OTC medications is equally problematic with teens and young adults abusing cough and cold medications. In a 2006 survey, 3.1 million persons aged 12 to 25 used an OTC cough and cold medication to get high.[4]

Research estimates that between 20% and 26% of primary care patients suffer from a substance use disorder.[5] The indicated prevalence must be considered in light of the fact that most of those with a substance use disorder do not seek out treatment in the substance abuse treatment system. The result is that the primary care physician becomes the first potential provider to recognize the problem and begin to address the issue with the patient. The importance of doing so cannot be trivialized because these disorders consistently rank among the 10 leading preventable risk factors for years of life lost to death and disability.[6] In addition, substance abuse disorders are associated with a higher risk of a variety of conditions, including hypertension, congestive heart failure, cirrhosis of the liver, lower back pain, arthritis, hepatitis C, pneumonia, chronic obstructive pulmonary disease, along with injuries and overdoses associated with misuse. Furthermore, substance use disorders are also responsible for the complication of many other medical conditions, contributing to the increasing cost associated with the health care system. Americans pay nearly $1000 annually to cover the costs of unnecessary health care, extra law enforcement, motor vehicle crashes, crime, and lost productivity due to substance abuse.[7]

Despite the impact illicit substances has on health care, physicians report low levels of preparedness to recognize and help patients with substance use disorders. Physicians cite a lack of adequate training in medical school, residency, and continuing education courses. Physicians also report skepticism about treatment effectiveness, time constraints, perceived patient resistance, discomfort with discussing substance abuse, and fear of losing patients. Patient self-report supports this information. In one study, 43% of patients said their physician never diagnosed their addiction, and 11% reported that the physician knew about their addiction, but did nothing about it.[8]

With this information in mind, the goal of this article is to briefly discuss the screening, diagnosis, management, and treatment of substance abuse, one of the more complicated and challenging public health issues faced by primary care physicians.

SCREENING/DIAGNOSIS

Despite modern medicine's many advances, detecting substance abuse problems has continued to challenge primary care providers. Screening is a vital first step, and many organizations, including the American Medical Association, recommend that clinicians routinely ask patients about substance use. Screening for a substance abuse problem relies heavily on asking the right questions. When the correct questions are asked, research has shown that the odds of successfully uncovering substance abuse in a general clinical setting can be improved significantly. There are numerous screening tools at the disposal of the primary care physician. In later discussion is a select sampling of the more popular measures used in the primary care setting.

Developed by the World Health Organization, the Alcohol Use Disorders Identification Test assesses alcohol consumption, drinking behaviors, and alcohol-related problems. This short, 10-item questionnaire is particularly suitable for use in primary care settings, has both a clinician-administered and a self-report version, and can been used with a variety of populations and cultural groups. The clinician-administered version should be administered by a health professional or paraprofessional.[9]

Modeled after the widely used Michigan Alcoholism Screening Test,[10] The Drug Abuse Screening Test (DAST) includes a 10-item (DAST-10) and a 20-item (DAST-20) brief screening tool to assess specifically for nonalcohol drug use. One of the fastest drug abuse screening tools around, each question requires a yes or no response and can be administered by a clinician or self-administered to both adults and older youth. This tool can be particularly useful in its ability to address the increasing epidemic of prescription drug abuse.

The CAGE-AID (*C*ut down, *A*nnoyed, *G*uilty, *E*ye-opener, *A*dapted to *I*nclude *D*rugs) questionnaire is a 4-item tool that takes 4 minutes or less to administer and is used to assess for drug/alcohol use problems in adults and adolescents (older than 16 years). The CAGE-AID questions focus on the behavioral effects of alcohol or drug use with higher scores indicating the need for the physician to ask more specific questions about frequency and quantity. This tool is good for health care professionals because it is easy to remember and can be administered by an interviewer, self-administered in pencil-and-paper format, or computer administered.

Knowing the type of agent involved in a substance use disorder is important for creating a management plan for the patient. Classes of substances used include stimulants, hallucinogens, and depressants. At times, the patient may only know a substance by street name. Many of the substances that are abused have their own profiles of symptoms but may share similar signs and symptoms with other drugs within the same class (**Table 1**).

To confirm the patient's drug of use, laboratory testing is a useful tool in screening patients for substance use disorders. However, in itself, it is not diagnostic. A multitude of modalities are used to screen patients for substance abuse, with the versatility of each modality increasing. Each one has their own advantages and disadvantages, including how easy it is to adulterate (**Table 2**). Urine testing is most widely used because of its availability, cost, and rapid results. It is helpful to keep in mind that common medications ingested by patients may cause false positives, and a correlation between the patient's history and physical diagnosis will help with diagnosing these.

Table 1
Profiles of signs and symptoms of intoxication and withdrawal of selected drugs

Substance	Intoxication	Withdrawal
Cocaine	Pupillary dilation, hypertension, cocaine bug hallucinations	"Crash" includes depression, fatigue, malaise, craving
Amphetamines	Pupillary dilation, hypertension, prolonged wakefulness	"Crash" includes depression, lethargy, stomach cramps, hunger
Alcohol	Loss of inhibition, labile moods, blackouts	Tachycardia, hypertension, delirium tremens
Opioids	Pinpoint pupils, nausea, vomiting, seizures	Anxiety, sweating, fever, diarrhea, yawning
Barbiturates/ benzodiazepines	Respiratory depression	Anxiety, seizures, tremors, insomnia
Phencyclidine (PCP)	Belligerence, fever, nystagmus, ataxia, fever	Depression, restlessness, thoughts and sleep disturbances
Marijuana	Euphoria, delusions, slowed time, increased appetite	Irritability, depression, nausea

Data from Kowalchuk A, Reed B. Substance use disorders. In: Rakel RE, Rakel D, editors. Textbook of family medicine. 9th edition. New York: Elsevier Saunders; 2016. p. 1159.e1; and Detoxification and Substance Abuse Treatment. No. 45. Center for Substance Abuse Treatment. Rockville (MD): Substance Abuse and Mental Health Services Administration (US); 2006.

Table 2
Strengths and weaknesses of different laboratory testing for substance abuse

Test	Tests Available	Advantages	Disadvantages
Breath	Alcohol	Immediate results	Blood levels needed for medical management, if needed
Blood	Most drugs	Difficult to manipulate, accurate, can detect recent ingestion	Drugs can clear quickly compared with urine; slightly invasive
Hair	Cocaine, marijuana, opiates, amphetamines, PCP	Noninvasive, detects patterns of use over time due to longest duration of positivity	Not widely available, drug or metabolite amounts may vary with hair color; difficult to process
Saliva	Amphetamines, alcohol, methamphetamines, opiates, cocaine, marijuana, PCP	Immediate results	Qualitative result only
Sweat (PharmChek sweat patch)	Cocaine, heroin, codeine, synthetic opiates, amphetamines, PCP, marijuana	Noninvasive, using patch	Quantification of drug levels difficult
Urine	Most drugs	Noninvasive, longer duration of positivity, rapid results, low cost	Easy to adulterate, collection is usually observed

Data from Kowalchuk A, Reed B. Substance use disorders. In: Rakel RE, Rakel D, editors. Textbook of family Medicine. 9th edition. New York: Elsevier Saunders; 2016. p. 1156; and Gwyther RE. Addiction. In: Essentials of family medicine. 3rd edition. New York; Baltimore (MD): Williams & Wilkins; 1993. p. 248.

The Diagnostic and Statistical Manual of Mental Disorders, 5th edition (DSM-V) criteria can be used to stratify severity of substance use disorders.[11] Characteristics of substance use, such as craving and desire for substances, impairment of daily activities, and frequency of use, are a part of the DSM-V criteria. Based on the severity, primary care providers can make appropriate decisions on how to manage community resources for the care of these patients. Important changes from the DSM-IV criteria include dropping use of the terms substance abuse and substance dependence, along with a removal of legal problems from qualifying criteria, replaced by the criteria of craving. DSM-IV and DSM-V criteria have been shown to have a good correlation for multiple substances.[12] However, it is yet to be seen whether the new DSM-V criteria will have an impact on management and outcomes.[13]

MANAGEMENT GOALS

Referral to a substance use clinic is recommended for those with substance use disorders.[14] However, primary care physicians are unique in their position to screen, treat, and refer patients misusing substances. Once detected, the management of these patients is paramount throughout the continuum of care and involves specific goals and the interventions by which to achieve these goals. Management goals should flow from the screening, which determined misuse, abuse, or dependence and contain specific elements. They include the following[5]:

- *Developing a treatment plan* that not only identifies the substance misuse but also targets associated problems such as physical conditions related to the misuse, and potential psychosocial aspects related to the problematic usage. Furthermore, the plan should contain specific goals and strategies to complete these objectives, along with the techniques and services to be provided by identified specialists.
- *Periodic reassessment to evaluate the patient's needs*, especially as it pertains to psychosocial or mental health issues. For example, suicidal ideation should be carefully monitored and dealt with promptly should the issue arise.
- *Ongoing monitoring of progress*, *as it relates not only to response to treatment but also to the patient's overall physical health*. This monitoring should be conducted through written notes that report detailed information regarding treatment effectiveness, ongoing laboratory tests, and so on. In addition, this should include follow-up to determine if work with the patient's therapist or counselor is helping as well as to gauge treatment compliance and participation.
- *Providing education* to help the patient understand their diagnosis, its cause, and prognosis as well as the risks involved with treatment. This education should include not only the patient but also others involved in his or her treatment, which may include family and/or spouse.

PHARMACOLOGIC STRATEGIES

Pharmacologic agents (**Table 3**) may play a major role in the treatment of substance abuse and depend on the setting and acuity in which an abused drug is encountered as well as the type of drug used. Immediate reversal of the drug may be needed in acute circumstances when respiratory or cardiac status may be threatened. Pharmacologic therapy used to intervene with long-term substance abuse takes many forms, including receptor agonists, receptor antagonists, or slow taper of an agent to wean without withdrawal symptoms. The goals of pharmacologic therapy are to reduce consumption and prevent relapse of the use of the drug. Many of these treatments are combined with nonpharmacologic interventions, which are discussed later.

Table 3
Examples of pharmacologic strategies

Substitution therapies	Methadone Buprenorphine Nicotine patches/gum
Blocking therapies	Naloxone Naltrexone
Triggered effect therapy	Disulfiram
Tapering therapy	Use with benzodiazepines

Data from Refs.[14–17]

Substitution Therapies

Substitution therapies include methadone and nicotine patches. With these therapies, the addictive agent is substituted in a different form. Methadone has improved the lives of many patients, but due to limited centers for distribution with resulting long waiting lists for required daily visits, patients may drive many miles to obtain this drug therapy. Buprenorphine is a partial agonist that comes in sublingual form by itself or in combination with naloxone, an opioid antagonist (Suboxone). Naloxone is only placed in the combination to prevent abuse of the therapy. Buprenorphine therapy has been proven to be safe and effective,[15,16] but can only be prescribed in office after training and receipt of a waiver from the Center for Substance Abuse Treatment and Drug Enforcement Administration.[14]

Another example of substitution therapy involves smoking cessation, wherein nicotine substitution is available in both transdermal patches and chewing gum. The patients are instructed to quit smoking on the day they start the treatment, with goal being reduction in the amount of nicotine intake over time through tapering doses.

Blocking Therapies

Blocking therapies are used quite frequently in acute and chronic cases of opiate abuse. Here, the medications are competitive agonists for the binding sites of the abused substances. Naloxone is used to reverse opiate overdose, whereas naltrexone is used in long-term use. Naltrexone is very popular for use in professionals who require monitoring for a licensing board. A strong motivator for naltrexone therapy is that relapse while on therapy can lead to losing a license to practice, leading to high success rates.[17]

Triggered Effect Therapies

Triggered effect therapies include disulfiram, which has been used in the past for alcohol abuse. By creating an accumulation of acetaldehyde, negative effects, such as nausea, vomiting, and vertigo, can be associated negatively with alcohol intake. Although previous literature had described disulfiram treatment falling out of favor, it recently has been shown that disulfiram may reduce drinking days and be effective at reducing chances of drinking in the short term.[18] Management with disulfiram requires patient compliance for effectiveness.

Tapering Therapy

Tapering therapy can be used with drugs such as benzodiazepines, where tapering is recommended over a several month period. This intervention is most successful in those who are highly motivated and can be trusted to frequently report back. Patients will be at risk of withdrawal symptoms, such as increased anxiety, depression, seizures, and altered mental status.[14]

NONPHARMACOLOGIC AND SELF-MANAGEMENT STRATEGIES

There is no single approach to treating and/or managing persons with a substance abuse disorder. Treatment strategies should be used while taking into consideration the diagnosis (misuse, abuse, or dependency/withdrawal) as well as any comorbidity. Moreover, equal importance must be given to addressing related social determinants of health associated with the substance abuse to improve outcomes, such as homelessness or unemployment. Before delving too deeply into specific treatment approaches, a brief discussion of the theoretic perspective as it relates to treatment is warranted. Current approaches reflect an integration of multiple models to address the problem. The following brief review can help primary care physicians understand the philosophic underpinnings of different treatment modalities, which include the following[5]:

1. The medical model: focuses on the physiologic causes of addiction. The emphasis is on the physician to use pharmacology as the primary means to alleviate symptoms and change behavior (methadone, for example, to treat withdrawal).
2. The psychological model: using a mental health professional, the emphasis is on emotional factors or learned patterns of maladaptive behavior (psychotherapy, for example).
3. The sociocultural model: changing the social, physical, and cultural environment to promote self-help through fellowship, spirituality, and social networks (Alcoholics Anonymous, for example).

Many times, a patient's journey to recovery and abstinence begins in the primary care physician's office. The first experience in identifying and addressing a patient's substance misuse is critical, due to not only its impact on treatment success but also the cost-conscious utilization of services while placing the patient in the least restrictive environment. As such, one of the tools that primary care physicians have at their disposal, which has demonstrated beneficial outcomes in the literature, is a brief intervention, which provides screening (CAGE questionnaire, for example), an assessment of the client's readiness to change, targeted intervention, and appropriate referral. The application of such a brief intervention model begins with a determination of hazardous, harmful, or dependent usage. Based on the results of the interview and the tool, primary care physicians are able to provide results, education, and recommendations to the patient to begin the change process. Below are listed the basic necessary components of a brief intervention[5]:

1. Give feedback about screening results, impairment, and risks while clarifying the findings.
2. Inform the patient about safe consumption limits and offer advice about change.
3. Assess the patient's readiness to change (see stages of change models).
4. Negotiate goals and strategies for change.
5. Arrange for follow-up treatment.
6. Referral for specialized treatment (if necessary).

Examples of brief interventions include the FRAMES Intervention Technique[19] (**Box 1**) and The Five "A's"[20] (**Box 2**). Brief interventions are best used with mild to moderate substance abuse as well as when apparent negative consequences due to current consumption are present. In addition, if comorbid conditions will be

Box 1
FRAMES intervention

Feedback about personal risk

Responsibility of the patient for change

Advice to change

Menu of strategies

Expression of empathy

Self-efficacy

Adapted from Miller WR, Sanchez VC. Motivating young adults for treatment and lifestyle change. In: Howard G, editor. Issues in alcohol use and misuse in young adults. Notre Dame (IN): University of Notre Dame Press; 1993.

exacerbated by substance misuse, or if the patient resists further outside assessment of recommended treatment, brief intervention within the primary care setting may be beneficial.[21] Another benefit of brief intervention is the ability to use it as a complement to, or in association with, ongoing specialized treatment. As the patient is able to demonstrate responsiveness, or lack thereof, to agreed on interventions, the treatment environment can be changed to more restrictive settings or can move in a less restrictive direction as indicated even if a more intensive treatment setting was initially agreed on. These settings include (from most intensive to least) inpatient hospitalization, residential treatment, intensive outpatient treatment, and outpatient treatment.

Inpatient hospitalization is the most restrictive and costly of these settings. Generally providing 24-hour around-the-clock care, this setting specializes in the medical management of detoxification using a multidisciplinary approach and is reserved typically for acute situations. Hospitalization is a short-term intensive approach lasting days to months and is used solely for the purpose of stabilizing the patient in order to move on to less restrictive care. Patients who have experienced a severe overdose, severe withdrawal symptoms, medical issues complicating withdrawal, a significant psychiatric comorbidity, or who are experiencing acute substance abuse and have not responded to other least restrictive forms of treatment are appropriate for this level of care.

Box 2
Five "A's" intervention (tobacco cessation example)

Ask: Identify and document tobacco use status for every patient at every visit.

Advise: In a clear, strong, and personalized manner, urge every tobacco user to quit.

Assess: Is the tobacco user willing to make a quit attempt at this time?

Assist: For the patient willing to make a quit attempt, use counseling and pharmacotherapy to help him or her quit.

Arrange: Schedule follow-up contact, in person, or by telephone, preferably within the first week after the quit date.

From Five Major Steps to Intervention (The "5 A's"). Agency for Healthcare Research and Quality, Rockville, MD. 2012. Available at: http://www.ahrq.gov/professionals/clinicians-providers/guidelines-recommendations/tobacco/5steps.html. Accessed October 1, 2015; with permission.

Residential treatment is the next level of care for patients who lack social support, who struggle with significant substance abuse problems, and who have tried other less restrictive environments and not been able to remain abstinent, but do not meet the criteria for hospitalization. These programs provide 24-hour supervision and are a live-in facility where treatment takes place via various modalities, including 12-step meetings, groups, individual therapy, and social skills training, along with treatment of psychological comorbidities. This particular modality tends to have better outcomes for certain patient populations, such as adolescents, for example, due in part to the change in environment.[22,23]

Intensive outpatient treatment, also known as partial hospitalization, can be the first transition from either a residential or an inpatient stay. This approach requires that the patient attend treatment typically during waking hours for 3 to 8 hours daily for up to 5 to 7 days per week. A full array of services is offered and is most suitable for patients with jobs and an extensive social network. Such patients may then transition to regular outpatient treatment whereby the same approaches are used, but in a more limited timeframe typically consisting of 1 to 2 visits one time weekly for 1 hour. This timeframe allows the patient to continue to focus on and reinforce treatment gains that were made in a more restrictive setting. Patients appropriate for this setting include those with significant motivation, transportation, and stable living arrangements. Treatment approaches used include group therapy and marital therapy, for example.

Increasingly, substance abuse/dependence is considered within an ongoing chronic disease framework with relapse always a possibility. As such, self-management for addictions is a vital treatment approach, encouraging the patient to take responsibility for their recovery by using resources within his or her environment and community.[24] Self-management allows the patient the autonomy to make treatment choices that will work for him or her with the goal of reducing misuse or maintaining abstinence. However, careful consideration must be made in regard to the individual patient's characteristics or circumstances because this approach is not for every patient. Such factors as unemployment, poverty, and lack of social support may make this a less desirable approach for patients experiencing these challenges. Involving oneself in Alcoholics Anonymous and other 12-step self-help and support group techniques are examples of this approach.

COMORBIDITIES AND RECURRENCE

Primary care clinicians need to be aware of comorbidities that are associated with substance use. Sexually transmitted diseases, such as human immunodeficiency virus, chlamydia, and gonorrhea, occur with increased frequency in patients with substance use.[13] Furthermore, intravenous drug users have their own set of comorbidities, including skin abscesses, cellulitis, infectious endocarditis, hepatitis, and pneumonia. Other mental health issues also run comorbid with substance use and must be addressed. In 2013, about 17.5% of adults (7.7 million) with any serious mental illness fit the criteria of substance use disorder.[1] Patients have an increased risk of a major depressive episode when using illicit drugs. In addition, clinicians should educate patients who are or will be considering pregnancy, because of the impact on fetal development many of these substances have.

Compliance with treatment plans for substance use disorders was found to be similar to other chronic medical illnesses, including hypertension, diabetes, and asthma.[25] The relapse rate of substance use disorders was also found to be similar to that of chronic illness, making ongoing support and monitoring with a primary care clinician crucial to successful treatment outcomes.[26]

SUMMARY

For primary care clinicians, substance abuse is an important aspect that must be addressed in taking care of the whole patient. With its relationship to other health conditions, substance use recognition and treatment are crucial to improved overall health for patients. Screening as well as anticipatory guidance during preventive visits can inform patients regarding the dangers of substance misuse/abuse, and treatment plans can integrate strategies to address these issues early on. With the multitude of screening tools, knowledge of the substances abused, and treatment resources and strategies, primary care physicians are well equipped to help their patients.

REFERENCES

1. Substance Abuse and Mental Health Services Administration. Results from the 2013 National Survey on Drug Use and Health: mental health findings. HHS Publication No. (SMA) 14–4887. Rockville (MD): Substance Abuse and Mental Health Services Administration; 2014.
2. Fact Sheets—Alcohol Use and Your Health. 2014. Available at: http://www.cdc.gov/alcohol/fact-sheets/alcohol-use.htm. Accessed October 1, 2015.
3. Substance Abuse and Mental Health Services Administration. Office of Applied Studies: results from the 2004 National Survey on Drug Use and Health: National Findings (NSDUH Series H-28, DHHS Publication No. SMA 05–4062). Rockville (MD): Department of Health and Human Services; 2005.
4. Alcohol, Tobacco, and Other Drugs. Substance Abuse and Mental Health Services Administration Web site. 2015. Available at: http://www.samhsa.gov/atod/other-drugs. Accessed November 1, 2015.
5. Center for Substance Abuse Treatment. A guide to substance abuse services for primary care clinicians. Treatment Improvement Protocol (TIP) Series, No. 24. Rockville (MD): Substance Abuse and Mental Health Services Administration (US); 1997.
6. Murray CJ, Atkinson C, Bjalla K, et al. The State of US Health, 1990-2010 burden of diseases, injuries, and risk factors. JAMA 2013;310:591–606.
7. Hall BP, Hawkinberry D, Moyers-Scott P. Prescription drug abuse & addiction: past, present and future: the paradigm for an epidemic. W V Med J 2010;106: 26–32.
8. CASA. The National Center on Substance Abuse at Columbia University. Missed opportunity: national survey of primary care physicians and patients on substance abuse. New York: National Center on Substance Abuse at Columbia University; 2000.
9. Screening and brief intervention for alcohol problems in primary health care. World Health Organization Web site. Available at: http://www.who.int/substance_abuse/activities/sbi/en/. Accessed October 1, 2015.
10. Selzer ML. The Michigan Alcoholism Screening Test (MAST): the quest for a new diagnostic instrument. Am J Psychiatry 1971;127:1653–8.
11. American Psychiatric Association. Diagnostic and statistical manual of mental disorders. 5th edition. Washington, DC: APA; 2013. Text revision (DSM-V).
12. Compton WM, Dawson DA, Goldstein RB, et al. Crosswalk between DSM-IV dependence and DSM-5 substance use disorders for opioids, cannabis, cocaine, and alcohol. Drug Alcohol Depend 2013;132:387–90.
13. Kowalchuk A, Reed B. Substance use disorders. In: Rakel RE, Rakel D, editors. Textbook of family medicine. 9th edition. New York: Elsevier Saunders; 2016. p. 1152–63.

14. Shapiro B, Coffa D, McCance-Katz EF. A primary care approach to substance misuse. Am Fam Physician 2013;88:113–21.
15. Ling W, Wesson DR. Clinical efficacy of buprenorphine: comparisons to methadone and placebo. Drug Alcohol Depend 2003;70(2 suppl):S49–57.
16. Mattick RP, Kimber J, Breen C, et al. Buprenorphine maintenance versus placebo or methadone maintenance for opioid dependence. Cochrane Database Syst Rev 2008;(2):CD002207.
17. Reading EG. Nine years' experience with chemically dependent physicians: the New Jersey experiences. Md Med J 1992;41(4):325–9.
18. Jorgensen CH, Pedersen B, Tonnesen H. The efficacy of disulfiram for the treatment of alcohol use disorder. Alcohol Clin Exp Res 2011;35:1749–58.
19. Miller WR, Sanchez VC. Motivating young adults for treatment and lifestyle change. In: Howard G, editor. Issues in alcohol use and misuse in young adults. Notre Dame (IN): University of Notre Dame Press; 1993. p. 55–81.
20. Five Major Steps to Intervention (The "5 A's"). Agency for Healthcare Research and Quality, Rockville, MD. 2012. Available at: http://www.ahrq.gov/professionals/clinicians-providers/guidelines-recommendations/tobacco/5steps.html. Accessed October 1, 2015.
21. Henry-Edwards S, Humeniuk R, Ali R, et al. Brief intervention for substance use: a manual for use in primary care. Geneva (Switzerland): World Health Organization; 2003.
22. Landry MJ. Overview of addiction treatment effectiveness. Pub. No. (SMA) 96–3081. Rockville (MD): Substance Abuse and Mental Health Services Administration; 1996.
23. American Psychiatric Association. Practice guidelines for treatment of patients with substance use disorders: alcohol, cocaine, opioids. Washington, DC: American Psychiatric Association; 1995.
24. Reardon C. Alternatives to 12-step addiction recovery. Social Work Today 2013;13:12.
25. McLellan AT, Lewis DC, O'Brien CP, et al. Drug dependence, chronic medical illness implications for treatment, insurance, and outcomes evaluation. JAMA 2000;284:1689–95.
26. Sdrulla AD, Chen G. The multidisciplinary approach to the management of substance abuse. In: Substance abuse inpatient and outpatient management. New York: Springer Science+Business Media; 2015. p. 193–208.